

Kinetic Theory of Decentralized Learning for Smart Active Matter

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Smart active matter has the ability to control its motion guided by individual policies to achieve collective goals. We introduce a theoretical framework to study a decentralized learning process in which agents can locally exchange policies to adapt their behavior and maximize a predefined reward function. We use our formalism to derive explicit hydrodynamic equations for the policy dynamics. We apply the theory to two different microscopic models where policies correspond either to fixed parameters similar to evolutionary dynamics or to state-dependent controllers known from the field of robotics. We find good agreement between theoretical predictions and agent-based simulations. By deriving fundamental control parameters and uncertainty relations, our work lays the foundations for a statistical physics analysis of decentralized learning.

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Decentralized optimization processes have been intensively studied in interdisciplinary fields using methods inspired from statistical physics, including the study of evolutionary and population dynamics in biology [1–4] and the investigation of social processes [5], such as opinion dynamics [6]. In addition, decentralized adaptation raises a growing interest in robotics and engineering [7–12], in particular to find more stable policies than typical centralized protocols [9,11]. Robustness and flexibility may be enhanced by drawing inspiration from collective animal behavior [13]. Standard experiments on robotic swarms typically involve around a hundred robots, and their numbers are increasing. This growth calls for a statistical physics approach to describe assemblies of small robots interacting through information exchange [11].

The idea of experimentally realizing large swarms of microrobots [14–16] as a way to build responsive or programmable metamaterials [17–20] is also at the core of the emerging field of smart active matter [21–27]. Along this soft robotics perspective, one tries to build large numbers of extremely simplified microrobots [28–31], whose size ranges from the granular scale [32,33] down to the colloidal one [20,34], and whose main features result from the physical properties of the soft material they are made of [35–38]. These soft microrobots may integrate simple computation capabilities [39], or may be controlled by an external computer through machine learning using, e.g., reinforcement learning techniques [40–49]. Although their starting points differ, both hard and soft robotics aim at building large assemblies of autonomous microrobots with communication and computation capabilities leading to adaptative collective behaviors [11,12,24,25,29,50,51]. This goal is also shared by the emerging topic of biological metamaterials, consisting for instance of insects aggregates [52,53].

The ongoing experimental and numerical development of smart active matter necessitates appropriate theoretical approaches to describe its collective properties. Early contributions in this direction include models of individual [54–58] or collective navigation of active particles [59,60], or of pedestrians in a dense crowd [61,62]. Stochastic thermodynamics may also be relevant to describe smart active matter [63,64]. Yet, apart from some first steps [63,65], a theoretical framework to describe the collective behavior of adaptive agents exchanging information with their neighbors [11,12] is still lacking.

In this Letter, we develop a kinetic theory framework to describe decentralized learning in smart active matter with local information exchange. In contrast to usual active matter [66–68], smart active matter consists of agents capable of responding to external stimuli, communicating and adapting their behavior. Through this adaptability, they can learn to solve specific tasks or to attain targeted collective properties. The behavior of individual agents is determined by a “policy,” i.e., a set of rules controlling their dynamics, whose parameters can be tuned. Among the modeling approaches used to characterize active matter, kinetic theories have played an important role by providing a framework to derive macroscopic hydrodynamic equations from microscopic collision rules [69–73]. We describe decentralized learning via the exchange of policies between neighboring agents, akin to collision rules, setting the stage for a kinetic theory framework, within which we systematically derive macroscopic hydrodynamic equations for the time evolution of policies starting from agent-based models.

Agent-based model of decentralized learning—We consider a generic system of N agents in d spatial dimensions representing, e.g., small robots [11]. Agents are located at position $\mathbf{x}$, move at velocity $\mathbf{v}$ with an overdamped

dynamics, and diffuse with a positional diffusion coefficient D_0 . Agents sense their environment (e.g., external fields or properties of neighboring particles) to control their dynamical rules. The set of adjustable parameters controlling the dynamics, whose values are specific to each agent, is here generically called the agent's policy $\mathcal{P}$. In the examples below, the policy either corresponds to the agent's angular diffusion coefficient, or to the response of the self-propulsion speed to an external stimulus. The state $\mathcal{S}$ of an agent gathers internal physical variables (e.g., orientation, self-propulsion velocity, etc.) and sensors outputs such as measured light intensity. Agents can then analyze observed states and extract the processed information $G(\mathcal{S})$, which fluctuates on a characteristic time scale τ_G . The processed information is then filtered and stored in the "memory" $\mathcal{M}$ via the running average on the time scale $\tau_{\mathcal{M}}$,

$$\frac{d\mathcal{M}(t)}{dt} = -\frac{1}{\tau_{\mathcal{M}}}[\mathcal{M}(t) - G(\mathcal{S}(t))], \quad (1)$$

where t is time. The purpose of the memory is to extract the relevant signal from potentially noisy observations $G(\mathcal{S})$. This ensures that reward is not awarded for attaining one specific state but rather to promote longtime behavior [11]. Equation (1) is equivalent to a convolution of the processed information $G(\mathcal{S})$ by an exponential memory kernel of characteristic time scale $\tau_{\mathcal{M}}$. The quantities $\mathcal{S}$, $\mathcal{M}$, and $\mathcal{P}$ may be multicomponent vectors. To lighten the presentation of the formalism, we stick to the scalar case, but the multidimensional formalism is presented in Supplemental Material (SM) [74].

To enable learning, agents can exchange information about the efficiency of their individual policies. This efficiency is quantified by a scalar reward function $\mathcal{R}(\mathcal{M})$, which is evaluated solely based on each agent's individual memory $\mathcal{M}$. Agents will thus optimize their personal reward, but the reward function may be chosen to promote a specific target collective behavior (see examples below). Comparing its own reward with other agents enables each agent to learn improved policies (Fig. 1). Agents are therefore randomly selected with rate $\lambda_T = \tau_T^{-1}$, called the teaching rate (τ_T is the teaching time). Once selected, an agent i iterates over all neighbors n , chosen in random order within the interaction radius r_c . For each selected neighbor n , agent i becomes a teacher (and agent n the student) with a probability $p_T(\mathcal{R}_i, \mathcal{R}_n)$ given by

$$p_T(\mathcal{R}_i, \mathcal{R}_n) = \frac{1}{2}(1 + \tanh[\alpha_T(\mathcal{R}_i - \mathcal{R}_n)]), \quad (2)$$

where $\mathcal{R}_i = \mathcal{R}(\mathcal{M}_i)$ and $\mathcal{R}_n = \mathcal{R}(\mathcal{M}_n)$. Otherwise, agent n is the teacher and agent i is the student. In either case, the student updates its policy $\mathcal{P}_S$ and memory $\mathcal{M}_S$ by adapting the corresponding values of the teacher, i.e., $\mathcal{P}_S^{\text{new}} = \mathcal{P}_T$ and

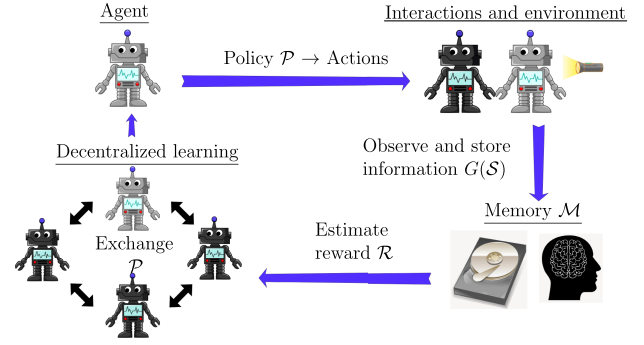


FIG. 1. Sketch of the decentralized learning procedure described in the Letter.

$\mathcal{M}_S^{\text{new}} = \mathcal{M}_T$. The memory of the student is updated together with the policy to make its reward consistent with the updated policy [11]. The parameter α_T in Eq. (2) controls the sharpness of p_T , and may be interpreted as the precision with which agents evaluate and communicate the reward $\mathcal{R}$. In Eq. (2) only reward differences matter and agents are lacking a concept of an absolute reward, which highlights the decentralized nature of the optimization. To enhance adaptability, we also include a small rate of spontaneous policy changes. By analogy with evolutionary processes in biology, we call these spontaneous changes "mutations" [75] and model them as a diffusion process in policy space with diffusion coefficient D_{mut} .

Kinetic theory of decentralized learning—To obtain a statistical description of the above agent-based model and derive the time evolution for the statistics of policies, we introduce a kinetic theory framework. This approach is natural since the learning events correspond to binary, intermittent interactions, which can be treated in analogy with binary collisions in usual kinetic theories. In this framework, an assembly of a large number N of agents is described by a single-agent phase-space density $f(\mathcal{S}, \mathcal{M}, \mathcal{P}, \mathbf{x}, t)$ characterizing the probability to find an agent at time t at position $\mathbf{x}$, in state $\mathcal{S}$, with memory $\mathcal{M}$ and policy $\mathcal{P}$. The time evolution formally reads

$$\frac{\partial f}{\partial t} = \mathcal{I}_{\text{phys}}[f] + \mathcal{I}_{\text{mem}}[f] + \mathcal{I}_{\text{learn}}[f]. \quad (3)$$

The term $\mathcal{I}_{\text{phys}}[f]$ describes the physical dynamics of the state $\mathcal{S}$ and the position $\mathbf{x}$, e.g., advection and diffusion, angular dynamics and physical interactions. $\mathcal{I}_{\text{mem}}[f]$ in Eq. (3) encodes the memory dynamics given in Eq. (1). Finally, the term $\mathcal{I}_{\text{learn}}[f]$ accounts for the dynamics of the policy $\mathcal{P}$, which includes diffusive mutations and communication-driven decentralized learning,

$$\mathcal{I}_{\text{learn}}[f] = D_{\text{mut}} \frac{\partial^2 f}{\partial \mathcal{P}^2} + \mathcal{I}_{\text{teach}}[f]. \quad (4)$$

The teaching term $\mathcal{I}_{\text{teach}}[f]$ describes the binary teaching events introduced in the agent-based model. Mathematically, the term $\mathcal{I}_{\text{teach}}[f]$ is a bilinear integral of the phase-space density f . The explicit, but lengthy, expression of $\mathcal{I}_{\text{teach}}[f]$ is reported in the End Matter (EM) section. We assume in this manuscript a time scale separation $\tau_G \ll \tau_M \ll \tau_T$, i.e., a hierarchy of timescales separating processed information dynamics τ_G , memory τ_M and teaching events τ_T . In other words, agents have a memory time τ_M long enough to filter the signal from noisy observations and are learning sufficiently slow to enable the memory to converge toward its steady-state value for a fixed policy. Information advection by particle motion therefore dominates over information propagation due to the finite communication range between agents. Within this approximation, our theory is thus purely local. However, it can be generalized to consider nonlocal communication or overlapping timescales.

To characterize the policy statistics, we assume that f is approximately Gaussian in $\mathcal{M}$ and $\mathcal{P}$, but this assumption can also be relaxed (see EM). Consequently, the reduced phase-space density

$$\varphi_0(\mathcal{P}, \mathbf{x}, t) = \int d\mathcal{S} \int d\mathcal{M} f(\mathcal{S}, \mathcal{M}, \mathcal{P}, \mathbf{x}, t) \quad (5)$$

can be described by its lowest-order moments. We also define the policy-dependent average memory,

$$\mu_{\mathcal{M}}(\mathcal{P}, \mathbf{x}, t) = \varphi_0^{-1} \int d\mathcal{S} \int d\mathcal{M} \mathcal{M} f(\mathcal{S}, \mathcal{M}, \mathcal{P}, \mathbf{x}, t). \quad (6)$$

The reduction steps from $f(\mathcal{S}, \mathcal{M}, \mathcal{P}, \mathbf{x}, t)$ to $\varphi_0(\mathcal{P}, \mathbf{x}, t)$ and $\mu_{\mathcal{M}}(\mathcal{P}, \mathbf{x}, t)$ are described in the EM. To derive closed hydrodynamic equations for the density and average policy, we expand $\mu_{\mathcal{M}}(\mathcal{P}, \mathbf{x}, t)$ around a predefined policy value $\mathcal{P}_*$. To ensure that this approximation is consistent, $\mathcal{P}_*$ should be a rough estimate of the optimal policy, yielding

$$\mu_{\mathcal{M}}(\mathcal{P}, \mathbf{x}, t) = \mu_{\mathcal{M}}^{(0)}(\mathbf{x}, t) + \mu_{\mathcal{M}}^{(1)}(\mathbf{x}, t)(\mathcal{P} - \mathcal{P}_*) + \dots \quad (7)$$

The quantities $\mu_{\mathcal{M}}^{(n)}(\mathbf{x}, t)$ are model specific, and can be determined explicitly under the timescale separation hypothesis [74]. We further introduce the agent density $\rho(\mathbf{x}, t) = \int d\mathcal{P} \varphi_0(\mathcal{P}, \mathbf{x}, t)$, the average policy $\mu(\mathbf{x}, t)$ and its variance, called *diversity*, $\sigma^2(\mathbf{x}, t)$, defined as

$$\mu(\mathbf{x}, t) = \rho^{-1} \int d\mathcal{P} \mathcal{P} \varphi_0(\mathcal{P}, \mathbf{x}, t), \quad (8)$$

$$\sigma^2(\mathbf{x}, t) = \rho^{-1} \int d\mathcal{P} (\mathcal{P} - \mu(\mathbf{x}, t))^2 \varphi_0(\mathcal{P}, \mathbf{x}, t). \quad (9)$$

For spatially uniform fields, $\rho(t)$, $\mu(t)$, and $\sigma^2(t)$, the evolution equations take the form

$$\frac{d\mu}{dt} = \rho \sigma^2 F_1(\mu, \sigma^2; \mu_{\mathcal{M}}^{(n)}), \quad (10)$$

$$\frac{d\sigma^2}{dt} = 2D_{\text{mut}} - \rho \sigma^4 F_2(\mu, \sigma^2; \mu_{\mathcal{M}}^{(n)}), \quad (11)$$

where $F_m(\mu, \sigma^2; \mu_{\mathcal{M}}^{(n)})$ with $m = 1, 2$ are model-specific polynomials [74] depending on the parameters $\mu_{\mathcal{M}}^{(n)}(t)$ introduced in Eq. (7), which can be determined independently. The closed Eqs. (10) and (11) for the policy dynamics are one of the major results of this work. The full hydrodynamic equations including space-dependent fields are derived for two specific models in SM [74]. An explicit model where space dependence is taken into account is given below. Importantly, we can already draw some nontrivial general conclusions from Eqs. (10) and (11). (i) The time derivative $\dot{\mu}(t)$ is proportional to the density $\rho(t)$, highlighting the many-body character of policy dynamics. Similarly, it is proportional to the diversity $\sigma^2(t)$, thus vanishing density or diversity imply vanishing adaptation rate $\dot{\mu}(t)$, consistent with Fisher's fundamental theorem of natural selection [76]. (ii) On time scales $t > \tau_M$ with a convex fitness function, we have $F_2(\mu(t), \sigma^2(t); \mu_{\mathcal{M}}^{(n)}(t)) > 0$, and thus $\sigma^2(t)$ decays monotonically in the absence of mutations. (iii) This emphasizes the importance of mutations D_{mut} to maintain a sufficient level of diversity and thus enable efficient optimization. Qualitatively, mutations therefore play a role similar to driving in athermal physical systems, as they break detailed balance and maintain a steady-state dynamics at long times.

Phototactic, diffusive agents—As a first application of the above framework, we study a model of phototactic microswimmers (in $d = 2$) with decentralized learning capabilities. Each swimmer moves at speed v_0 in a direction given by its orientation angle θ in the (x, y) plane, which defines the state $\mathcal{S} = \theta$. As for active Brownian particles [77], orientation can change randomly via rotational diffusion with strength D_θ , but also via tumbling events toward the x direction (direction of the light source) [78]. Physical interactions between microswimmers are neglected. The target behavior is a macroscopic flow along the x axis in which the ensemble-averaged velocity $\langle V_x \rangle = N^{-1} \sum_{i=1}^N V_{x,i}$ is equal to the target velocity V_T . This behavior can be promoted via a decentralized learning procedure in which each agent measures its own velocity $G(\mathcal{S}) = V_x = v_0 \cos \theta$ and adapts its rotational diffusion coefficient $\mathcal{P} = D_\theta$ (i.e., its policy). By maximizing the reward function, $\mathcal{R}(\mathcal{M}) = -(\mathcal{M} - V_T)^2$, each agent's time-averaged local velocity, $\mathcal{M}$, converges toward V_T and thus induces the same ensemble-averaged velocity.

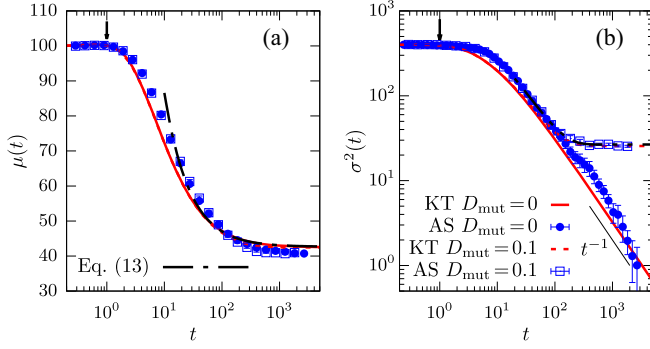


FIG. 2. Adaptation of the microswimmer model to a constant target velocity $V_T = 0.85$: Results of kinetic theory (KT) and agent-based simulations (AS) without mutations ($D_{\text{mut}} = 0$) and with mutations ($D_{\text{mut}} = 0.1$). The vertical arrow indicates $\tau_M = 1$. (a) Average policy (rotation diffusion) $\mu(t)$. (b) Diversity $\sigma^2(t)$ (variance of policy). The black dashed-dotted curve is Eq. (12) fitted to the AS data with $D_{\text{mut}} = 0.1$.

Importantly, although physical interactions are neglected, learning is collective due to information exchange. The asymptotic solutions of Eqs. (10) and (11) (detailed derivations in [74]) read,

$$\sigma^2(t) = \sqrt{\frac{2D_{\text{mut}}}{\lambda_0}} \frac{1}{\tanh[\sqrt{2D_{\text{mut}}\lambda_0}(\tau_0 + t)]} \quad (12)$$

$$\mu(t) = D_{\theta,T} + \frac{\eta_0 \sqrt{2D_{\text{mut}}\lambda_0}}{\sinh[\sqrt{2D_{\text{mut}}\lambda_0}(\tau_0 + t)]}, \quad (13)$$

which boil down to $\sigma^2(t) = [\lambda_0(\tau_0 + t)]^{-1}$ and $\mu(t) = D_{\theta,T} + \eta_0/(\tau_0 + t)$ for $D_{\text{mut}} \rightarrow 0$. The parameter $\lambda_0 = 4\lambda_T \alpha_T \rho V_*'^2$ is a characteristic rate, where $V_*' = \partial \bar{V}_x(D_\theta)/\partial D_\theta|_{D_\theta=D_{\theta,*}}$ is the response of the average velocity to policy changes and $\bar{V}_x(D_\theta) = \langle V_x \rangle_{D_\theta}$ is the ensemble-averaged flow for fixed policy D_θ . The parameters τ_0 and η_0 depend on initial conditions, and $D_{\theta,*} = \mathcal{P}_*$ is the predefined policy used in Eq. (7). When $D_{\text{mut}} > 0$, $\sigma^2(t)$ converges to a plateau value $\sigma_\infty^2 = \sqrt{2D_{\text{mut}}/\lambda_0}$ when $t \rightarrow \infty$, while $\mu(t)$ converges to the optimal policy $D_{\theta,T}$. The convergence of both $\sigma^2(t)$ and $\mu(t)$ occurs over the learning time $\tau_L = (2D_{\text{mut}}\lambda_0)^{-1/2}$. Importantly, our theoretical analysis reveals two important control parameters: the rate λ_0 and the mutation strength D_{mut} . The effect of the mutation rate D_{mut} is characterized by the “uncertainty relation” $\tau_L \sigma_\infty^2 = \lambda_0^{-1}$, where both τ_L and σ_∞^2 depend on D_{mut} , while λ_0 is independent of D_{mut} . The uncertainty relation states that quick convergence and adaptability (i.e., a small τ_L) obtained by increasing the mutation rate D_{mut} comes at the cost of having large fluctuations around the target policy (i.e., a large σ_∞^2), in turn implying fluctuations around the collective goal, since $\text{Var}(N^{-1} \sum_{i=1}^N V_{x,i}) \propto \sigma_\infty^2/N$. Predictions of the kinetic

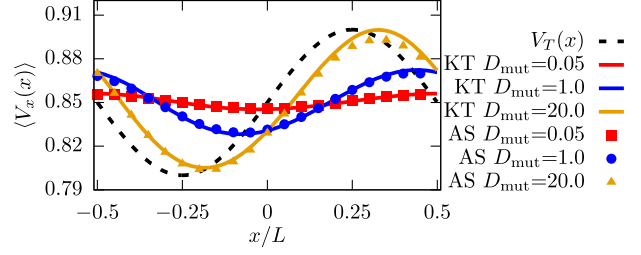


FIG. 3. Adaptation of the microswimmer model to a space-dependent target $V_T(x) = 0.85 + 0.05 \sin(2\pi x/L)$ (black line). Results obtained by kinetic theory (KT) and agent-based simulations (AS) are shown for different mutation rates D_{mut} .

theory are successfully compared to agent-based simulations in Fig. 2. Additionally, Eq. (12) can be used to fit the longtime behavior of $\sigma^2(t)$ on agent-based simulations data (see Fig. 2) to determine λ_0 and D_{mut} , as on experimental data where such parameters are unknown. We extract $D_{\text{mut}}^{\text{fit}} = 0.094 \pm 0.002$ and $\lambda_0^{\text{fit}} = (2.65 \pm 0.01) \times 10^{-4}$, which compare well to the agent-based model values $D_{\text{mut}} = 0.1$ and $\lambda_0 = 2.655 \times 10^{-4}$.

We now consider a space-dependent target velocity $V_T(x) = V_T^s + V_T^A \sin(2\pi x/L)$ in a system of size L with periodic boundary conditions. The reward function thus also becomes space dependent, $\mathcal{R}(\mathcal{M}, x) = -(\mathcal{M} - V_T(x))^2$. At leading order, the steady-state average policy $\mu(x)$ satisfies [74],

$$-V_T^s \nabla \mu(x) - \sqrt{2D_{\text{mut}}\lambda_0}(\mu(x) - D_{\theta,T}(x)) = 0, \quad (14)$$

where $D_{\theta,T}(x)$ is the optimal policy maximizing the local reward. We find for the average local velocity

$$\langle V_x(x) \rangle = V_T^s + V_x^A(\phi) \sin(2\pi x/L - \phi), \quad (15)$$

with a phase ϕ given by $\tan(\phi) = 2\pi V_T^s / (\sqrt{2D_{\text{mut}}\lambda_0}L)$ and an amplitude $V_x^A(\phi) = V_T^A \cos(\phi)$. The theory thus predicts that agents with smaller mutation rates have a larger phase ϕ and therefore the observed deviation from the target profile $\langle V_x(x) \rangle - V_T(x) = V_T^A(\cos(\phi) \sin(2\pi x/L - \phi) - \sin(2\pi x/L))$ increases. In Fig. 3, we compare this predicted behavior by kinetic theory to agent-based simulations and find very good agreement for both the mutation-rate dependent amplitude $V_x^A(D_{\text{mut}})$ and the phase $\phi(D_{\text{mut}})$. The observed phenomenology can be explained by comparing the typical advection time $\tau_A = L/V_T^s$ to the learning time τ_L . Agents with small mutation rates have $\tau_L \gg \tau_A$ thus they learn too slowly and are unable to adapt in time to the environment. Our results therefore emphasize the importance of mutations to maintain diversity and thus being able to adapt to space- or time-dependent targets.

Light-sensing robots—We also study a model featuring robots (in $d = 1$) which attempt to maximize light collected from a heterogeneous light field $I(x)$. The state of a

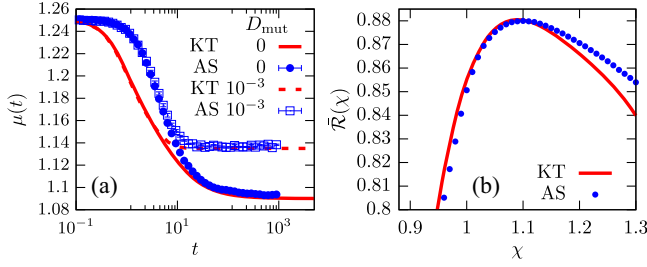


FIG. 4. Adaptation of the light-sensing robot model to maximize collected light: kinetic theory (KT) and agent-based simulations (AS) without mutations ($D_{\text{mut}} = 0$) and with mutations ($D_{\text{mut}} = 10^{-3}$). (a) Average policy $\mu(t) = \langle \chi \rangle$; (b) average reward $\bar{\mathcal{R}}(\chi)$ for robots with a given policy χ .

robot is $\mathcal{S} = (I, v)$, where I is the local light intensity measured by a sensor, and v is the robot velocity (diffusion is neglected, $D_0 = 0$). The memory $\mathcal{M}$ averages the measured light intensity according to Eq. (1), with $G(\mathcal{S}) = I$. A state-dependent controller adapts the robot's speed $v(I) = v_0 - \chi I$, keeping $v(I) \geq v_{\text{min}}$. The parameter χ thus defines the sensitivity with which robots react to the external light. Robots adapt their policy $\mathcal{P} = \chi$ to maximize collected light, using a reward $\mathcal{R}(\mathcal{M}) = \mathcal{M}$. Figure 4(a) shows that the theory successfully describes the optimization of χ , by predicting the longtime behavior of $\mu(t)$ (calculations in SM [74]). In the absence of mutations ($D_{\text{mut}} = 0$), the diversity $\sigma^2(t)$ goes to zero and $\mu(t)$ approaches the maximal reward around $\chi_T = 1.09$ [see Fig. 4(b)]. However, for $D_{\text{mut}} > 0$, mutations cause the robots to learn a distribution of policies with $\sigma_\infty^2 > 0$. Importantly, the underlying model implies a pronounced asymmetry $\bar{\mathcal{R}}(\chi_T + \epsilon) > \bar{\mathcal{R}}(\chi_T - \epsilon)$, $\epsilon > 0$ in the reward function [Fig. 4(b)]. Here, $\bar{\mathcal{R}}(\chi) := \mathcal{R}[\mu_{\mathcal{M}}(\chi)]$ is the average steady-state reward for an agent with policy χ , which we can derive exactly since the assumed timescale separation implies the existence of a direct relationship between the policy and the steady-state average memory, $\mu_{\mathcal{M}}(\chi)$. This asymmetry makes it preferable to adapt for $t \rightarrow \infty$ to $\mu_\infty > \chi_T$ in the presence of mutations, as observed in the agent-based simulations. Expanding $\mu_{\mathcal{M}}$ to fourth order in Eq. (7), KT provides a quantitative prediction of μ_∞ [Fig. 4(a)]. The increased steepness of $\bar{\mathcal{R}}(\chi)$ for $\chi > 1.2$ in KT implies a larger $V'_* = \partial \bar{\mathcal{R}}(\chi) / \partial \chi$ and thus explains the faster adaptation speed observed for KT in Fig. 4(a).

Conclusion and outlook—We proposed a kinetic theory framework for decentralized learning in smart active matter, in which agents maximize their reward $\mathcal{R}$ by learning improved policies $\mathcal{P}$ from neighboring agents, to reach a target collective behavior. Our theory enables derivation of space-dependent field equations describing the time-dependence of policy distributions. Assumptions required to derive closed equations include weak correlations between agents, a hierarchy of timescales, a Gaussian

approximation for the $\mathcal{M}$ and $\mathcal{P}$ dependence, and expansions of the teaching probability p_T and the average memory. Most of these assumptions can be lifted for specific applications; however, the theory cannot quantitatively account for many-body interactions and correlations. We have applied the framework to two explicit models, and predictions favorably compared to numerical simulations of the agent-based model, showing a rich phenomenology. Future research may include the analysis of learning with different interaction ranges, multidimensional or discrete policies, and more complex reward functions such as time-dependent or nonmonotonic functions or multiple species with competing reward. More generally, it will be crucial to explore systematic techniques guiding the design of reward functions facilitating targeted collective behavior. Addressing this inverse problem may be possible by incorporating higher-level learning mechanisms which are able to iteratively optimize the agent-based reward function itself. While we considered for simplicity a limit of separated timescales between physical, memory, and learning dynamics, our framework allows for systematic improvements over this approximation, notably taking into account the effect of the communication range. In addition, it would also be of interest to investigate the generality of the uncertainty relation linking learning time and policy fluctuations.

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Data availability—The data that support the findings of this Letter are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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End Matter

We describe here the general framework of kinetic theory of decentralized learning introduced in this Letter. Calculation details for specific models are reported in SM [74].

The term $\mathcal{I}_{\text{teach}}[f]$, introduced in Eq. (4), can be formulated in a rather similar way as usual collision terms in kinetic theories. It is convenient to express $\mathcal{I}_{\text{teach}}[f]$ in a similar way as a master equation for a stochastic jump process, using an f -dependent effective transition rate $W_f(\mathcal{M}', \mathcal{P}' | \mathcal{M}, \mathcal{P}; \mathbf{x}, t)$ defined as

$$\begin{aligned} W_f(\mathcal{M}', \mathcal{P}' | \mathcal{M}, \mathcal{P}; \mathbf{x}, t) \\ = 2\lambda_T \int d\mathcal{S}_2 \int d\mathcal{M}_2 \int d\mathcal{P}_2 \int d\mathbf{x}_2 K(\mathbf{x}_2, \mathbf{x}) \\ \times p_T(\mathcal{R}(\mathcal{M}_2), \mathcal{R}(\mathcal{M})) f(\mathcal{S}_2, \mathcal{M}_2, \mathcal{P}_2, \mathbf{x}_2, t) \\ \times \delta(\mathcal{P}' - \mathcal{P}_2) \delta(\mathcal{M}' - \mathcal{M}_2). \end{aligned} \quad (\text{A1})$$

The transition rate W_f encodes the microscopic learning dynamics of the agent model: the probability that a focus particle 1, at position $\mathbf{x}$ and characterized by $(\mathcal{S}, \mathcal{M}, \mathcal{P})$, learns its memory $\mathcal{M}' = \mathcal{M}_2$ and policy $\mathcal{P}' = \mathcal{P}_2$ from a particle 2 at position $\mathbf{x}_2$ and with $(\mathcal{S}_2, \mathcal{M}_2, \mathcal{P}_2)$, is proportional to the teaching rate λ_T , the distance-dependent kernel $K(\mathbf{x}_2, \mathbf{x})$, and to the teaching probability $p_T(\mathcal{R}(\mathcal{M}_2), \mathcal{R}(\mathcal{M}))$ (i.e., the probability that particle 2 becomes the teacher of particle 1), defined in Eq. (2). The kernel $K(\mathbf{x}_2, \mathbf{x})$ has a characteristic range δ_c , e.g., $K(\mathbf{x}_2, \mathbf{x}) = (2\pi\delta_c^2)^{-1} \exp[-(\mathbf{x}_2 - \mathbf{x})^2/2\delta_c^2]$ in 2D, and is normalized to 1. Note that if the agent learns the policy $\mathcal{P}_2$ it also adapts its memory to the value $\mathcal{M}_2$ in order to evaluate the correct reward associated with policy $\mathcal{P}_2$. This choice is motivated by Ref. [11], which similarly exchanged the information about the reward function sampled by the robots to avoid additional longtime sampling. By replacing $\delta(\mathcal{M}' - \mathcal{M}_2) \rightarrow \delta(\mathcal{M}' - \mathcal{M})$, the student would maintain their previous memory, which might be a more intuitive choice in other applications. The state $\mathcal{S}$ of particle 1 is not included in $W_f(\mathcal{M}', \mathcal{P}' | \mathcal{M}, \mathcal{P}; \mathbf{x}, t)$ because $\mathcal{S}$ is not modified by the learning process. The factor of 2 in front of λ_T accounts for the fact that during one adaptation step either agent 1 is the teacher and agent 2 the student or the other way around.

Note also that at variance with usual kinetic theories (e.g., of gases, granular gases, or active matter), the

occurrence of learning interactions is not controlled by the kinetics of particle collisions, but rather by an imposed teaching rate λ_T , which in practice may be much smaller than the rate of particle encounter. We thus assume that information propagation due to the sole communication range, at a speed at most at the order of $\lambda_T \delta_c$, remains much slower than information transport by particle motion at a typical speed v_0 (i.e., we assume $\lambda_T \delta_c \ll v_0$). This means that we can neglect the distinction between $\mathbf{x}_2$ and $\mathbf{x}$ in Eq. (A1), thereby yielding a purely local expression of W_f ,

$$W_f(\mathcal{M}', \mathcal{P}' | \mathcal{M}, \mathcal{P}) = 2\lambda_T p_T(\mathcal{R}(\mathcal{M}'), \mathcal{R}(\mathcal{M})) f_0(\mathcal{M}', \mathcal{P}'), \quad (\text{A2})$$

where we have introduced the marginal distribution

$$f_0(\mathcal{M}, \mathcal{P}) = \int d\mathcal{S} f(\mathcal{S}, \mathcal{M}, \mathcal{P}). \quad (\text{A3})$$

Here, and in the following, we drop the explicit dependence on $\mathbf{x}$ and t for the sake of brevity. Within kinetic theory the teaching term $\mathcal{I}_{\text{teach}}[f]$ is formally obtained from an effective master equation, and takes the form

$$\begin{aligned} \mathcal{I}_{\text{teach}}[f] = \int d\mathcal{M}' \int d\mathcal{P}' [W_f(\mathcal{M}, \mathcal{P} | \mathcal{M}', \mathcal{P}') f(\mathcal{S}, \mathcal{M}', \mathcal{P}') \\ - W_f(\mathcal{M}', \mathcal{P}' | \mathcal{M}, \mathcal{P}) f(\mathcal{S}, \mathcal{M}, \mathcal{P})]. \end{aligned} \quad (\text{A4})$$

By inserting the explicit form of the transition rate in Eq. (A2), the teaching term can be written down explicitly,

$$\begin{aligned} \mathcal{I}_{\text{teach}}[f] = 2\lambda_T f_0(\mathcal{M}, \mathcal{P}) \int d\mathcal{M}' p_T(\mathcal{R}, \mathcal{R}') \\ \times \int d\mathcal{P}' f(\mathcal{S}, \mathcal{M}', \mathcal{P}') - 2\lambda_T f(\mathcal{S}, \mathcal{M}, \mathcal{P}) \\ \times \int d\mathcal{M}' p_T(\mathcal{R}', \mathcal{R}) \int d\mathcal{P}' f_0(\mathcal{M}', \mathcal{P}'), \end{aligned} \quad (\text{A5})$$

with the shorthand notations $\mathcal{R} = \mathcal{R}(\mathcal{M})$ and $\mathcal{R}' = \mathcal{R}(\mathcal{M}')$. Writing decentralized learning in such a general way and thus introducing a framework for the systematic derivation of macroscopic hydrodynamic equations for population dynamics is one of the major results of this Letter. As in usual kinetic theories, we have assumed that

the two-body phase-space density can be approximated by multiplying the one-body phase-space densities, thus assuming weak correlations between individual agents. In the following, we now aim at transforming these equations to derive macroscopic equations for the dynamics of policies.

In a first reduction step, we integrate out the dependence of f on the state $\mathcal{S}$ and memory $\mathcal{M}$ to focus on the macroscopic dynamics of policies. As a first approximation this can be achieved by assuming $\tau_G \ll \tau_M \ll \tau_T$, i.e., that the time scale τ_M is large enough to smooth out the fluctuations in the observations of $G(\mathcal{S})$ on a timescale τ_G , and that the learning occurs on time scales much larger than changes in the memory. This assumption also implies that exchanging information on the memory during learning does not impact the reported results. We believe this assumption corresponds to a natural learning scenario, allowing each agent to sample as much information as possible before learning. However, the assumption of a hierarchy in time scales is not a fundamental limitation of the theory and can also be lifted if this is desirable for specific applications.

When studying the time dependence of f on a time scale τ_M , the physical interaction terms in $\mathcal{I}_{\text{phys}}[f]$ drop out (leaving only the diffusive and convective terms). Transforming the agent-based memory dynamics in Eq. (1) into an evolution equation for the probability distribution of $\mathcal{M}$ [79] leads to the explicit relation for the memory dynamics,

$$I_{\text{mem}}[f] = \tau_M^{-1} \left(f(\mathcal{S}, \mathcal{M}, \mathcal{P}) + (\mathcal{M} - G(\mathcal{S})) \frac{\partial f(\mathcal{S}, \mathcal{M}, \mathcal{P})}{\partial \mathcal{M}} \right), \quad (\text{A6})$$

we find a closed equation for the time dependence of the marginal distribution,

$$\begin{aligned} \frac{\partial f_0(\mathcal{M}, \mathcal{P})}{\partial t} = & -\nabla J_0 + \tau_M^{-1} \left(f_0 + (\mathcal{M} - \bar{G}(\mathcal{P})) \frac{\partial f_0}{\partial \mathcal{M}} \right) \\ & + 2\lambda_T f_0 \int d\mathcal{M}' \tanh(\alpha_T(\mathcal{R} - \mathcal{R}')) \\ & \times \int d\mathcal{P}' f_0(\mathcal{M}', \mathcal{P}') + D_{\text{mut}} \frac{\partial^2}{\partial \mathcal{P}^2} f_0, \end{aligned} \quad (\text{A7})$$

with the current $J_0 = (\bar{V}(\mathcal{P}) - D_0 \nabla) f_0$. At this level, due to the timescale separation, the details of the microscopic model thus only enter via the policy-dependent average of the observed information $\bar{G}(\mathcal{P}) := \langle G(\mathcal{S}) \rangle_{\mathcal{P}}$, and the policy-dependent average velocity $\bar{V}(\mathcal{P}) := \langle V \rangle_{\mathcal{P}}$. The brackets $\langle \dots \rangle_{\mathcal{P}}$ denote the policy-dependent ensemble average used as a proxy for the running average on a timescale τ_M due to the assumed timescale separation.

As in the main text, we then introduce the marginal phase-space distribution of policies [Eq. (5)], the agent

density $\rho(\mathbf{x}, t) = \int d\mathcal{P} \varphi_0(\mathcal{P}, \mathbf{x}, t)$ and the average memory, $\mu_{\mathcal{M}}(\mathcal{P}, \mathbf{x}, t) = \varphi_0^{-1} \int d\mathcal{M} \mathcal{M} f_0(\mathcal{M}, \mathcal{P}, \mathbf{x}, t)$ [Eq. (6)]. Due to the assumed timescale separation a Gaussian dependence of f_0 on $\mathcal{M}$ can be justified with the central limit theorem. To find closed equations we approximate the activation function to first order, $\tanh(\alpha_T(\mathcal{R} - \mathcal{R}')) = \alpha_T(\mathcal{R} - \mathcal{R}') + \mathcal{O}(\alpha_T^3)$. This limits applications of the theory to cases where it is not necessary to evaluate the reward with very high precision, and learning is achieved as long as sufficient teaching events are performed. After some calculations we find,

$$\begin{aligned} \frac{\partial \varphi_0(\mathcal{P})}{\partial t} = & -\nabla J + D_{\text{mut}} \frac{\partial^2}{\partial \mathcal{P}^2} \varphi_0(\mathcal{P}) \\ & + \tilde{\lambda}_T \rho \int d\mathcal{M} f_0(\mathcal{M}, \mathcal{P}) \mathcal{R}(\mathcal{M}) \\ & - \tilde{\lambda}_T \varphi_0(\mathcal{P}) \int d\mathcal{P}' \int d\mathcal{M}' f_0(\mathcal{M}', \mathcal{P}') \mathcal{R}(\mathcal{M}'), \end{aligned} \quad (\text{A8})$$

with the policy current $J = (\bar{V}(\mathcal{P}) - D_0 \nabla) \varphi_0$. We also introduce the renormalized teaching rate, $\tilde{\lambda}_T = 2\lambda_T \alpha_T$. For the average memory we can derive

$$\frac{\partial \mu_{\mathcal{M}}(\mathcal{P})}{\partial t} = \tau_M^{-1} (\bar{G}(\mathcal{P}) - \mu_{\mathcal{M}}(\mathcal{P})). \quad (\text{A9})$$

This equation would also include terms proportional to D_0 , D_{mut} and $\tilde{\lambda}_T$, which, however, are negligible compared to τ_M^{-1} within the approximations applied in the present manuscript.

Subsequently, we perform the reduction procedure as described in the main text to derive closed equations for the average policy $\mu(t)$ and the diversity $\sigma^2(t)$, as presented in Eq. (10). The closure is based on the assumption of a Gaussian dependence of the reduced phase-space density on $\mathcal{P}$. This assumption is reasonable for the examples discussed in the present Letter featuring an unbounded policy space and convex fitness functions. (The phototactic agents had policies $\mu(t) \gg \sigma(t)$; thus, this bound did not intervene with the results.) If Gaussianity cannot be assumed, for example, in applications in which the policy space is bounded, other distributions such as the β distribution or multimodal distributions may be considered.

While we have focused on continuous quantities within this Letter, our theory can certainly be adapted to accommodate discrete quantities. This includes, in particular, discrete policies which might be relevant for decision-making. Let $\mathcal{P}_0$ be the set of possible discrete policies, and $f_{\mathcal{P}_0}(\mathcal{M}, \mathbf{x}, t)$ be the phase-space density of agents with policy $\mathcal{P} \in \mathcal{P}_0$. Then Eq. (A7) becomes

$$\begin{aligned}
 \frac{\partial f_{\mathcal{P},0}(\mathcal{M})}{\partial t} = & -\nabla J_0 + \tau_{\mathcal{M}}^{-1} \left(f_0 + (\mathcal{M} - \bar{G}_{\mathcal{P}}) \frac{\partial f_{\mathcal{P},0}}{\partial \mathcal{M}} \right) \\
 & + 2\lambda_T f_0 \int d\mathcal{M}' \tanh(\alpha_T(\mathcal{R} - \mathcal{R}')) \\
 & \times \sum_{\mathcal{P}' \in \mathcal{P}_0} f_{\mathcal{P}',0}(\mathcal{M}') + \sum_{\mathcal{P}' \in \mathcal{P}_0} \lambda_{\text{mut}, \mathcal{P}' \rightarrow \mathcal{P}} f_{\mathcal{P}',0}.
 \end{aligned}
 \tag{A10}$$

The only differences with Eq. (A7) are the replacement of the integral by a sum, and introducing a discrete master equation describing mutation via a random jump process in policy space.